8-4-2026

UNIT 2

Correlation Analysis

Measuring extent of relationship between 2 variables.

Types of correlation

1. Positive correlation: The values of variables deviate in same direction.

   eg: Advertising V/s sales revenue

   Income and expenditure.

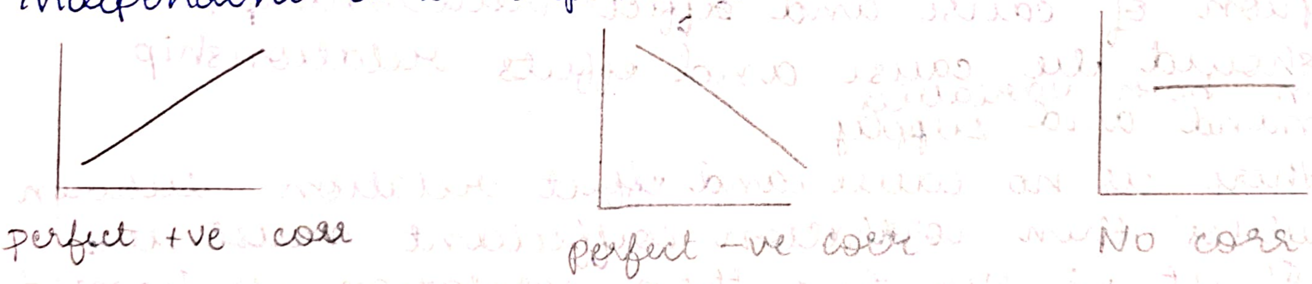
2. Negative correlation: The values of variables deviate in opposite direction.

   eg: Literacy and Poverty.

   Price and demand of commodity

Scatter plot / Scatter diagram:

It is a statistical tool to determine correlation between independent and dependent variable.



Linear and non linear correlation

The correlation between 2 variables is linear if corresponding unit change in one variable and there is constant change in other variable.

~~B~~ Correlation between 2 variables is non linear if corresponding unit change in one variable then other variable doesn't change at constant rate.

Methods of studying correlation analysis

1. Karl Pearson's correlation coefficient.
2. Spearman Rank corelation.

correlation coefficient

$$r = \frac{\Sigma dx dy}{\sqrt{(\Sigma dx)^2 (\Sigma dy)^2}}$$

$dx = x - \bar{x}$
$dy = y - \bar{y}$
mean

0 — no correlat
+1 - perfect +ve
-1 - perfect -ve
>0.75 - strong +ve
<0.25 - weak +ve

>-0.75 strong -ve
<-0.25 weak -ve

** List & explain

Assumptions for corelation coefficient

1) Assumption of linearity: The variables used to know the corelation coefficient must be linearly related and can be seen through scatterplot.

2) Assumption of normality: Both the variables under the study should follow normal distribution.

** any distribution following 68:95:99 ratio of would be normally distributed.

3) Assumption of cause and effects relationship: There should be cause and effects relationship between both variables
eg: demand and supply.
when there is no cause and effect relation between variables then correlation coefficiant must be zero. If it is non zero then corelation is termed as chance correlation.
eg: Rainfall and litracy in state over a time period.

Q1. Calculate Karl Pearson's coeff of corelation between expenditure on advertizing and sales from the data given below.

Advertising expenses	39	65	62	90	82	75	25	98	36	78
Sales	47	53	58	86	62	68	60	91	51	84

X	Y	$dx = x - \bar{x}$	$dy = y - \bar{y}$	$(dx)^2$	$(dy)^2$	$dxdy$
39	47	-26	-19	676	361	494
65	53	0	-13	0	169	0
62	58	-3	-8	9	64	24
90	86	25	20	625	400	500
82	62	17	-4	289	16	-68
75	68	10	2	100	4	20
25	60	-40	-6	1600	36	240
98	91	33	25	1089	625	825
36	51	-29	-15	841	225	435
78	84	13	18	169	324	234
650	660			Σ 5398	2224	2704

$$\bar{X} = \frac{650}{10} = 65$$

$$\bar{Y} = \frac{660}{10} = 66$$

$$r = \frac{\Sigma dxdy}{\sqrt{\Sigma(dx)^2 \, \Sigma(dy)^2}} = \frac{2704}{\sqrt{5398 \times 2224}} = 0.7804$$

Strong positive corelation.

Properties of corelation coffeciation

1 Correlation coffecient can't exceed 1 numerically in other words it lies between -1 to +1

2 Correlation coeff is independent of change of origin & scale, x and y are the given variable transformed into new variable u and v

$$u = \frac{x-A}{h} \qquad v = \frac{y-B}{k} \qquad h, k > 0$$

Two independent variables are un-correlated by the converse is not true.

Rank Correlation

$$\rho = 1 - \frac{6 \Sigma d^2}{n(n^2-1)}$$

Pearson's correlation between the ranks x and y is called Rank correlation coeff b/w the characteristic a & b. where b is the difference b/w pair of ranks of the same individual in 2 characteristics & n is the no. of pairs.

$d_i = x_i - y_i$

1. Calculate the spearmen rank coefficient b/w advertisement cost & sales from the following data

Adv cost	39	65	62	90	82	75	25	98	36	78
sales	47	53	58	86	62	68	60	91	51	84

$n = 10$
no. of observations

x	y	R_x	R_y	$d = R_x - R_y$	d^2
39	47	8	10	-2	4
65	53	6	8	-2	4
62	58	7	7	0	0
90	86	2	2	0	0
82	62	3	5	-2	4
75	68	5	4	1	1
25	60	10	6	4	16
98	91	1	1	0	0
36	51	9	9	0	0
78	84	4	3	1	1

$\Sigma d^2 = 30$

$$\rho = 1 - \frac{6\Sigma d^2}{n(n^2-1)}$$

$$= 1 - \frac{6(30)}{10(100-1)}$$

$$= 1 - \frac{180}{990}$$

$$= \frac{9}{11} = 0.818$$

strong correlation

Regression Analysis

Regression analysis is a mathematical ~~method~~ measure of the average relationship b/w 2 or more variables and the variable used for our is independent variable and the var. whose value is predicted by independent variable is the dependent variable.

Lines of Regression

The lines of the best fit that express the average relationships b/w the variables

The lines of regression provide the best estimate to the value of dependent variable for a specific value of independent variable

Regression analysis when cofined to study of only 2 variables at a time is called as simple regression and Regression analysis for studying more than 2 variables at a time is called as multiple regression

b_{yx} = Coeff of Regression of y on x

b_{xy} = Coeff of Regression of x on y

lines of regression

$r^2 = b_{yx} \cdot b_{xy}$

$y - \bar{y} = b_{yx}(x - \bar{x})$

$(x - \bar{x}) = b_{xy}(y - \bar{y})$

$$b_{yx} = \frac{r\sigma_x}{\sigma_x} \qquad b_{xy} = \frac{r\sigma_x}{\sigma_y}$$

$$b_{yx} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2}$$

$$b_{xy} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(y-\bar{y})^2}$$

1. From the following data obtain 2 regression equation

Sales	91	97	108	121	67	124	51	73	~~10~~ 111	57
Purchase	71	75	69	97	70	91	39	61	80	47

$\bar{x} = 90$

$\bar{y} = 70$

x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$	$(x-\bar{x})^2$	$(y-\bar{y})^2$
91	71	1	1	1	1	1
97	75	7	5	35	49	25
108	69	18	-1	-18	324	1
121	97	31	+27	837	961	729
67	70	-23	0	0	529	0
124	91	34	21	714	1156	441
51	89	-39	-31	1209	1521	961
73	61	-17	-9	153	289	81
111	80	21	10	210	441	100
57	47	-33	-23	759	1089	529
				3900	6360	2868

$$b_{yx} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2} = \frac{3900}{6360} = 0.6132$$

$$b_{xy} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(y-\bar{y})^2} = \frac{3900}{2868} = 1.3598$$

$$y - 70 = 0.6132(x-90)$$

$$x - 90 = 1.3598(y-70)$$

$y - 70 = 0.6132(x-90)$

$y - 70 = 0.6132x - 55.188$

$0.6132x - y = 55.188 - 70$

$\boxed{0.6132x - y + 14.812 = 0}$

$x - 90 = 1.3598y - 95.186$

$\boxed{x - 1.3598y + 5.186 = 0}$

$r^2 = b_{yx} \cdot b_{xy}$

$r^2 = 0.8338$

$r = 0.91314$

Angle between 2 Regression Lines.

If θ is the acute angle between 2 regression lines in case of 2 variables x and y.

$$\tan\theta = \frac{1-r^2}{r}\left[\frac{\sigma_x \sigma_y}{\sigma_x^2+\sigma_y^2}\right]$$

case 1 = If $r = 0$

$\tan\theta = \infty \Rightarrow \theta = \frac{\pi}{2}$

case 2: If $r = \pm 1$

$\tan\theta = 0 \Rightarrow \theta = 0$ or π

1. If θ is angle between 2 regression lines and SD of y is twice the SD of x and $r = 0.25$, $\tan\theta = ?$

$$\tan\theta = \frac{1-r^2}{r}\left[\frac{\sigma_x \sigma_y}{\sigma_x^2+\sigma_y^2}\right] = \frac{1-(0.25)^2}{0.25}\left[\frac{\sigma_x 2\sigma_x}{\sigma_x^2+4\sigma_x^2}\right]$$

$$= \frac{1-(0.25)^2}{0.25}\left[\frac{2\cancel{\sigma_x^2}}{5\cancel{\sigma_x^2}}\right] = \frac{3}{2}$$

$\theta = 56.30^\circ = 0.982$ rad

Partial correlation:

It is the correlation between 2 variables after removing the linear effects of other variable on them.

The correlation coefficient between x_1 and x_2 after eliminating linear ~~algebra~~ effects of x_3 on x_1 and x_2 is called as partial correlation coefficient

$$r_{12.3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{(1-r_{13}^2)(1-r_{23}^2)}}$$

$$r_{13.2} = \frac{r_{13} - r_{12} r_{32}}{\sqrt{(1-r_{12}^2)(1-r_{32}^2)}}$$

$$r_{23.1} = \frac{r_{23} - r_{21} r_{31}}{\sqrt{(1-r_{21}^2)(1-r_{31}^2)}}$$

Total correlation between 2 variables

$$r_{xy} = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2] - [n\sum y^2 - (\sum y)^2]}}$$

1. From the following data obtain $r_{12.3}$, $r_{13.2}$, $r_{23.1}$

x_1	x_2	x_3	x_1^2	x_2^2	x_3^2	x_1x_2	x_1x_3	x_2x_3
20	12	13	400	144	169	240	260	156
15	13	15	225	169	225	195	225	195
25	16	12	625	256	144	400	300	192
26	15	16	676	225	256	390	416	240
28	23	14	784	529	196	644	392	322
40	15	18	1600	225	324	600	720	270
38	28	14	1444	784	196	1064	532	392
192	122	102	5754	2332	1510	3533	2845	1767

$$r_{12} = \frac{n\sum x_1x_2 - (\sum x_1)(\sum x_2)}{\sqrt{[n\sum x_1^2 - (\sum x_1)^2][n\sum x_2^2 - (\sum x_2)^2]}}$$

$$= \frac{7(3533) - (192)(122)}{\sqrt{[7(5754) - (192)^2][7(2332) - (122)^2]}} = \frac{1307}{\sqrt{(3414) - 1440}}$$

0.58947

$$r_{13} = \frac{n\sum x_1x_3 - (\sum x_1)(\sum x_3)}{\sqrt{[n\sum x_1^2 - (\sum x_1)^2][n\sum x_2^2 - (\sum x_2)^2]}} = \frac{7(2845) - (192)(102)}{[7(5754) - (192)^2][7(2332) - (122)^2]}$$

$=$

$= 0.4396$

$$r_{23} = \frac{n\sum x_2x_3 - (\sum x_2)(\sum x_3)}{\sqrt{[n\sum x_2^2 - (\sum x_2)^2][n\sum x_3^2 - (\sum x_3)^2]}} = \frac{7(1767) - (122)(102)}{\sqrt{(7(2332) - (122)^2)(7(1510) - (102)^2)}}$$

$r_{23} = -0.1534$

$$r_{12.3} = \frac{r_{12} - r_{13} r_{23}}{\sqrt{(1-r_{13}^2)(1-r_{23}^2)}} = \frac{0.58947 - (0.4396)(-0.1534)}{\sqrt{(1-0.4396^2)(1-(0.1534)^2)}}$$

$$= 0.7228$$

$$r_{13.2} = \frac{r_{13} - r_{12} r_{32}}{\sqrt{(1-r_{12}^2)(1-r_{32}^2)}} = \frac{0.4396 - (0.58947)(-0.1534)}{\sqrt{(1-0.58947^2)(1-(-0.1534)^2)}}$$

$$= 0.664$$

$$r_{23.1} = \frac{r_{23} - r_{21} r_{31}}{\sqrt{(1-r_{21}^2)(1-r_{31}^2)}} = \frac{(-0.1534) - (0.58947)(0.4396)}{\sqrt{(1-0.58947^2)(1-0.4396^2)}}$$

$$= -0.56857$$

Multiple corelation

It is the study ~~correlation~~ of combined influence of 2 or more variables on a single variable x_1, x_2, x_3 are the variables having observations on n indivisuals or limits.

Multiple correlation coefficient of x_1 on x_2 and x_3 is a simple correlation coefficient b/w x_1 and jointed effect of x_2 and x_3. It can also be defined as correlation between x_1 and its estimate based on x_2 & x_3. Multiple corellation coefficient is a simple correlation coefficient between variable & its estimates.

$$R_{1.23}^2 = \frac{r_{12}^2 + r_{13}^2 - 2r_{12} r_{13} r_{23}}{1 - r_{23}^2}$$

$$R_{2.13}^2 = \frac{r_{12}^2 + r_{23}^2 - 2r_{12} r_{13} r_{23}}{1 - r_{13}^2}$$

$$R_{3.12}^2 = \frac{r_{13}^2 + r_{23}^2 - 2r_{12} r_{13} r_{23}}{1 - r_{12}^2}$$

1. Calculate $R_{1\cdot 23}$, $R_{3\cdot 12}$, $R_{2\cdot 13}$

X_1	X_2	X_3	X_1^2	X_2^2	X_3^2	X_1X_2	X_2X_3	X_1X_3
2	3	1	4	9	1	6	3	2
5	6	3	25	36	9	30	18	15
7	10	6	49	100	36	70	60	42
11	12	10	121	144	100	132	120	110
25	31	20	199	289	146	238	201	169

$$r_{12} = \frac{n\Sigma x_1 x_2 - (\Sigma x_1)(\Sigma x_2)}{\sqrt{(n\Sigma x^2 - (\Sigma x)^2]\,[n\Sigma x_2^2 - (\Sigma x_2)^2]}}$$

$$= \frac{4(238) - (25)(31)}{\sqrt{[4(199) - (25)^2]\,[4(289) - (31)^2]}} = 0.969$$

$$r_{13} = \frac{n\Sigma x_1 x_3 - (\Sigma x_1)(\Sigma x_3)}{\sqrt{(n\Sigma x_1^2 - (\Sigma x_1)^2]\,[n\Sigma x_3^2 - (\Sigma x_3)^2]}} = \frac{4(169) - (25)(20)}{\sqrt{(4(199) - 25^2)\,(4(146) - 20^2)}}$$

$$= 0.992$$

$$r_{23} = \frac{4(201) - (31)(20)}{\sqrt{[4(289) - 31^2]\,[4(146) - 20^2)}} = 0.9713$$

$$R^2_{1.23} = \frac{r_{12}^2 + r_{13}^2 - 2r_{12}r_{13}r_{23}}{1 - r_{23}^2} = \frac{0.969^2 + 0.992^2 - 2(0.969)(0.992)(0.9713)}{1 - (0.9713)^2}$$

$$= 0.98459$$

$R_{1.23} = 0.992$

$R^2_{2.13} = 0.944$

$R_{2.13} = 0.971$

$R^2_{3.12} = 0.985$

$R_{3.12} = 0.992$

2. The heights of father and son in inches are given. Find 2 lines of regression, one for predicting the height of the son based on father height and another for predicting height of father based on the son's height. Estimate the average height of a son when the height of father is 68.5

X height of father	65	66	67	67	68	69	70	71
Y height of son	66	68	65	69	74	73	72	70

X	Y	$(X-\bar{x})$	$(Y-\bar{y})$	$(X-\bar{x})^2$	$(Y-\bar{y})^2$	$(X-\bar{x})(Y-\bar{y})$
65	66	-2.875	-3.625	+8.266	13.141	10.422
66	68	-1.875	-1.625	+3.516	2.641	3.047
67	65	-0.875	-4.625	+0.766	21.391	4.047
67	69	-0.875	-0.625	+0.766	0.391	0.547
68	74	0.125	4.375	0.016	19.141	0.547
69	73	1.125	3.375	1.266	11.391	3.797
70	72	2.125	2.375	4.516	5.641	5.047
71	70	3.125	0.375	9.766	0.141	1.172
543	557			28.878	73.878	28.626

$\bar{X} = 67.875$

$\bar{Y} = 69.625$

$$b_{yx} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(x-\bar{x})^2} = \frac{28.626}{28.878} = 0.99$$

$$b_{xy} = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{\Sigma(y-\bar{y})^2} = \frac{28.626}{73.878} = 0.387$$

$r^2 = b_{yx} \times b_{xy} = 0.383 \qquad r = 0.619$

$y-\bar{y} = b_{yx}(x-\bar{x})$

~~$y - 67.875 = (0.99)(x - 69.62)$~~

$y - 69.625 = (0.99)(x - 67.875)$

$y - 69.625 = 0.99x - 67.19625$

$\boxed{0.99x - y + 2.428 = 0}$

$x-\bar{x} = b_{xy}(y-\bar{y})$

$x - 67.875 = 0.387(y - 69.625)$

$x - 67.875 = 0.387y - 26.945$

$\boxed{x - 0.387y + \cancel{94.82}\ 40.93 = 0}$

$0.99x - y + 2.428 = 0$

$0.99(68.5) - y + 2.428 = 0$

$y = 0.99(68.5) + 2.428 = 70.243$

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3. A medium sized company has been investing in advertising to bost its overall profit. The marketing team has recorded their advertisement expenditure and corresponding profit figures over 6 different months. They now wish to analyze whether there is a significant relationship between how much they spend in advertising & profit they earn. Calculate the correlation b/w expenditure & profit. Interpret the results.

~~month Advertisr~~

months	1	2	3	4	5	6
advertisement expenditure	30	44	45	43	34	44
Profit	56	55	60	64	62	63

X	Y	$dx = x - \bar{x}$	$dy = y - \bar{y}$	$(dx)^2$	$(dy)^2$	$dxdy$
30	56	-10	-4	100	16	40
44	55	+4	-5	16	25	-20
45	60	+5	0	25	0	0
43	64	+3	4	9	16	12
34	62	-6	2	36	4	-12
44	63	4	3	16	9	12
240	360			202	70	32

$\bar{X} = \frac{240}{6} = 40$ $\quad \bar{Y} = 60$

$r = \frac{\Sigma dxdy}{\sqrt{(\Sigma dx)^2 (\Sigma dy)^2}}$

$= \frac{32}{\sqrt{202 \times 70}} = 0.269$

∴ positive corelation